Sahaj Bhadja

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Summary

Software engineer with 2+ years building, testing, debugging, and supporting database-backed applications using Java, Python, JavaScript, SQL, object-oriented design, REST APIs, automated testing, performance tuning, CI/CD, and technical documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant **Azure SQL, SQL Server, and Blob Storage** services for 500+ users, tuning schemas and indexed queries to reduce application dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant FastAPI workflows for 1,000+ monthly jobs using explicit states, validation, error handling, and failure isolation, tripling throughput while improving production debugging, maintenance, and reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Python REST APIs and React workflows handling 10,000+ daily requests, integrating secure file ingestion and AWS RDS persistence while maintaining stable application performance under production workloads.
- Designed multithreaded batch processing for 1,000+ files with automated validation, fault-tolerant execution, and failure recovery, reducing turnaround by 60% while strengthening software testing and defect resolution.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform microservices and 20+ **PyTest and Mockito** suites through Jenkins CI/CD, automating unit, integration, and regression validation across Windows, Linux, and macOS production environments.
- Optimized REST API and ETL components using object-oriented patterns, AWS Glue, S3, and SQL, reducing response times by 30% and improving distributed data-processing efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a modular application with REST APIs, WebSockets, and eight Node.js services, supporting 2M+ synchronized records while translating requirements into maintainable real-time business workflows.
- Automated Docker-based builds, testing, and deployments through Jenkins CI/CD, reducing release time by 50% while improving code quality, documentation, and defect resolution across frequent production releases.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built Flask backend APIs for a booking application supporting reservations, scheduling, status updates, and payments across 50,000+ records while maintaining synchronized relational and Firebase-backed application data.
- Implemented webhook-based payment synchronization and troubleshooting workflows using Flask and Firebase, reducing manual payment handling by 70% while improving operational visibility, error recovery, and customer support responsiveness.

Skills

Languages & Applications: Java, Python, JavaScript, TypeScript, SQL, C++, Bash, object-oriented programming, design patterns, REST APIs, JSON, FastAPI, Flask, Node.js, React, WebSockets, microservices; **Databases & Performance:** SQL Server, Azure SQL, PostgreSQL, MySQL, Oracle, MongoDB, Firebase, AWS RDS, relational databases, NoSQL, schema design, indexing, query tuning, data validation, ETL, performance tuning; **Quality & Operations:** Windows, Linux, AWS, Azure, Docker, Jenkins, Git, CI/CD, PyTest, Mockito, unit testing, integration testing, regression testing, debugging, defect resolution, error handling, automation, monitoring, application support, ticket triage, SDLC, Agile/Scrum, code review, technical documentation

Projects

Chat2SQL Code | *Java, TypeScript, SQL Server, Oracle, Docker*
- Built an AI-assisted Java and TypeScript database workspace supporting 10+ relational and NoSQL systems, accelerating SQL authoring, schema exploration, validation, and application data management across platforms.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, Algorithms*
- Implemented six scheduling algorithms with dynamic metrics and Gantt visualization, validating preemptive and non-preemptive execution across 1,000+ process combinations while reducing application debugging time by 20%.

HMIS Live | *TypeScript, MERN, REST APIs, Sentry, Twilio*
- Built a database-backed workflow application integrating consent-based APIs, appointment scheduling, automated alerts, and Sentry monitoring, reducing administrative workload by 30% across secure user and staff operations.